Data Science Capstone Project Report

SpaceX Falcon 9 First-Stage Landing Prediction

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GitHub Repository:

github.com/ankitarchoudhary/IBM-Data-Science-Capstone

Executive Summary

90

Launches Analyzed

66.7%

Overall Success Rate

94.4%

Best Model Accuracy

Methods Used

- Data collection: SpaceX REST API + Wikipedia web scraping
- Data wrangling: derived binary landing outcome label
- EDA: Matplotlib/Seaborn visualization + SQL queries
- Interactive analytics: Folium map + Plotly Dash dashboard
- Predictive modeling: 4 classification algorithms, tuned with GridSearchCV

Key Results

- KSC LC-39A: highest site success rate (77.3%)
- GTO orbit: most common, but lowest success rate (51.9%)
- Success rate rose from 0% (2013) to ~90% (2019)
- KNN model: best predictor at 94.4% test accuracy
- Zero false negatives in the best model's test predictions

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Introduction

Project Background

SpaceX advertises Falcon 9 launches at a fraction of the cost charged by other providers, largely because it reuses the first stage instead of discarding it after every flight. If the outcome of a landing attempt can be predicted from launch conditions, that prediction can be used to estimate launch cost. This project collects, explores, and models real Falcon 9 launch data to make that prediction.

Problem Statement

- How do payload mass, launch site, and orbit affect first-stage landing success?
- Has the landing success rate improved over time?
- Which classification algorithm predicts landing outcome most accurately?

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Data Collection - SpaceX API

Structured JSON launch data retrieved directly from the SpaceX REST API.

- GET request to the /launches endpoint
- Parse JSON response into a DataFrame with pd.json_normalize()
- Resolve rocket, launchpad, payload, and core IDs via follow-up API calls
- Filter to single-core Falcon 9 launches only
- Fill missing PayloadMass values with the column mean
- Export to dataset_part_1.csv

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Data Collection - Web Scraping

Supplementary launch data scraped from the Wikipedia list of Falcon 9 and Falcon Heavy launches.

- Request a fixed, archived Wikipedia revision (for reproducibility) with BeautifulSoup
- Locate the launch records table among all tables on the page
- Extract clean column names from table headers
- Loop through table rows, parsing date, booster, site, payload, orbit, customer, and outcome
- Use regex to reliably split combined date-time text into separate Date and Time fields
- Export to dataset_part_2.csv

Data Wrangling Methodology

The raw Outcome column (e.g. "True ASDS", "False Ocean") combines landing success and landing type as text. This was converted into a clean binary target label for modeling.

Class = 0 (Unsuccessful)

- False ASDS / False RTLS / False Ocean - landing attempted but failed
- None None - no core landing data available
- None ASDS - drone-ship landing planned, outcome unrecorded

Class = 1 (Successful)

- True ASDS - successful drone-ship landing
- True RTLS - successful return-to-launch-site landing
- True Ocean - successful controlled ocean landing

EDA with Data Visualization - Overview

Matplotlib and Seaborn were used to visualize relationships between launch conditions and landing outcome, in order to identify which features are most predictive before modeling.

- **Scatter: Flight Number vs. Launch Site** — Shows whether success became more common over time, and whether that differs by site.
- **Scatter: Payload Mass vs. Launch Site** — Shows how payload mass and outcome interact at each site.
- **Bar Chart: Success Rate by Orbit** — Compares landing reliability across the 11 orbit types in the data.
- **Line Chart: Yearly Success Trend** — Tracks how SpaceX's landing success rate evolved from 2010 to 2020.

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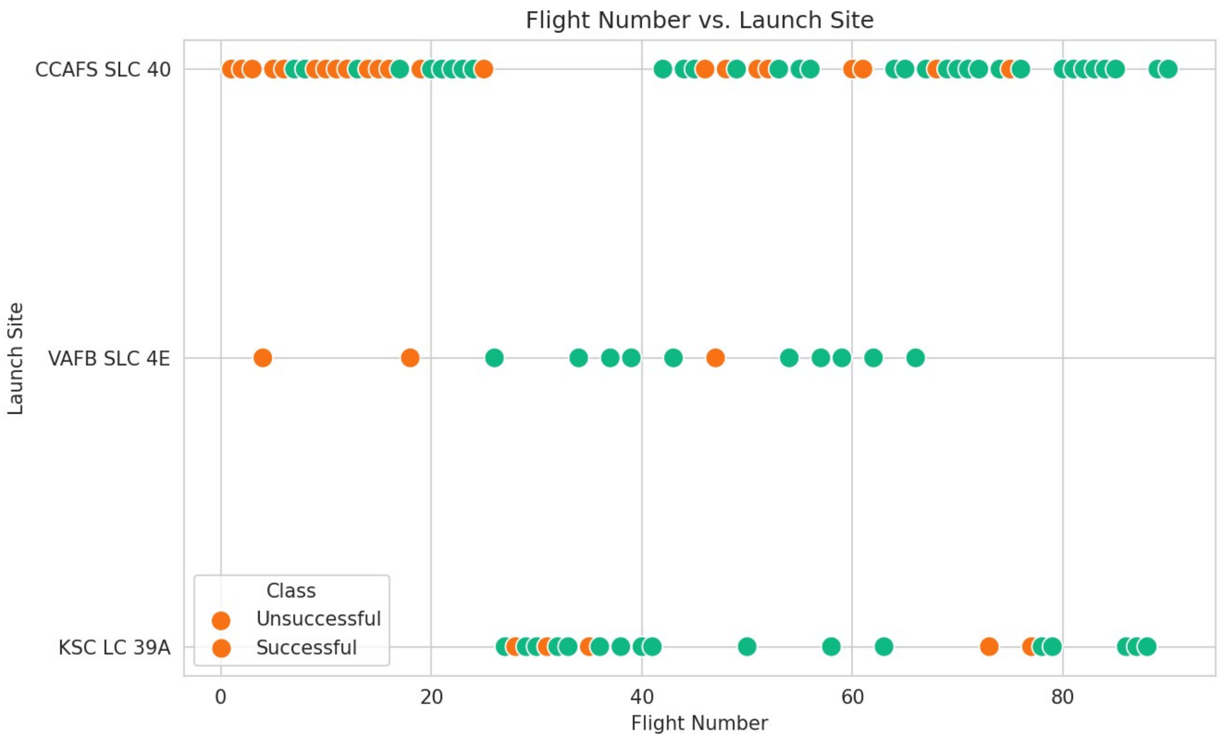
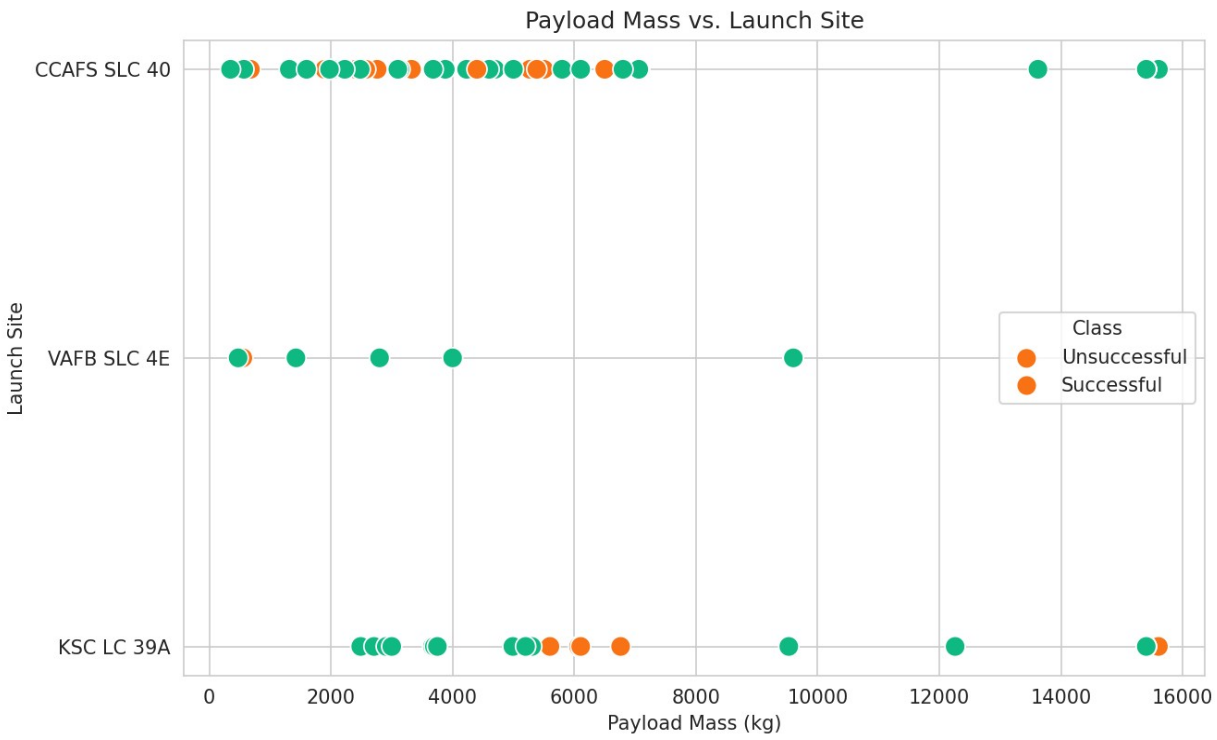
Result: Flight Number vs. Launch Site



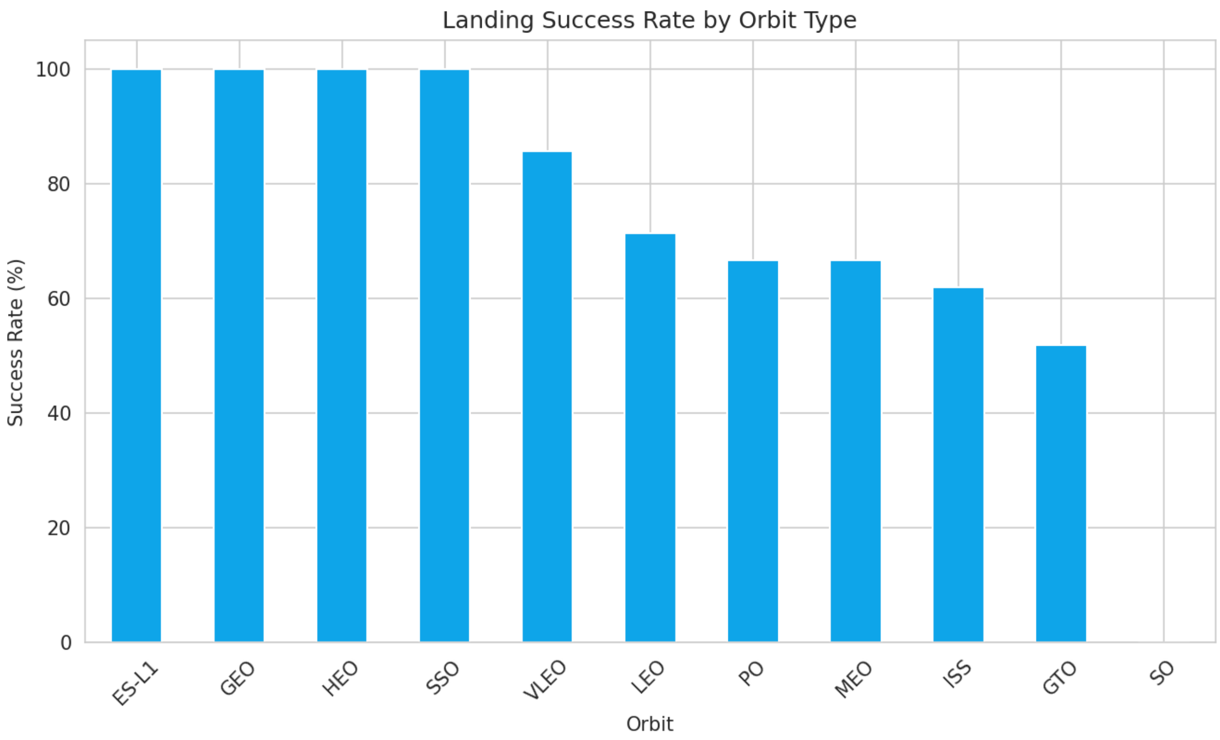
- Analysis: Early flight numbers (roughly 1-20) show mostly unsuccessful landings across all three sites, while later flight numbers are dominated by successful outcomes.
- This confirms that landing success is strongly correlated with time (flight number), consistent with SpaceX's improving landing technology.
- CCAFS SLC-40 accounts for the largest share of launches across the full flight-number range, while KSC LC-39A only begins appearing from the mid-20s onward.

Result: Payload Mass vs. Launch Site



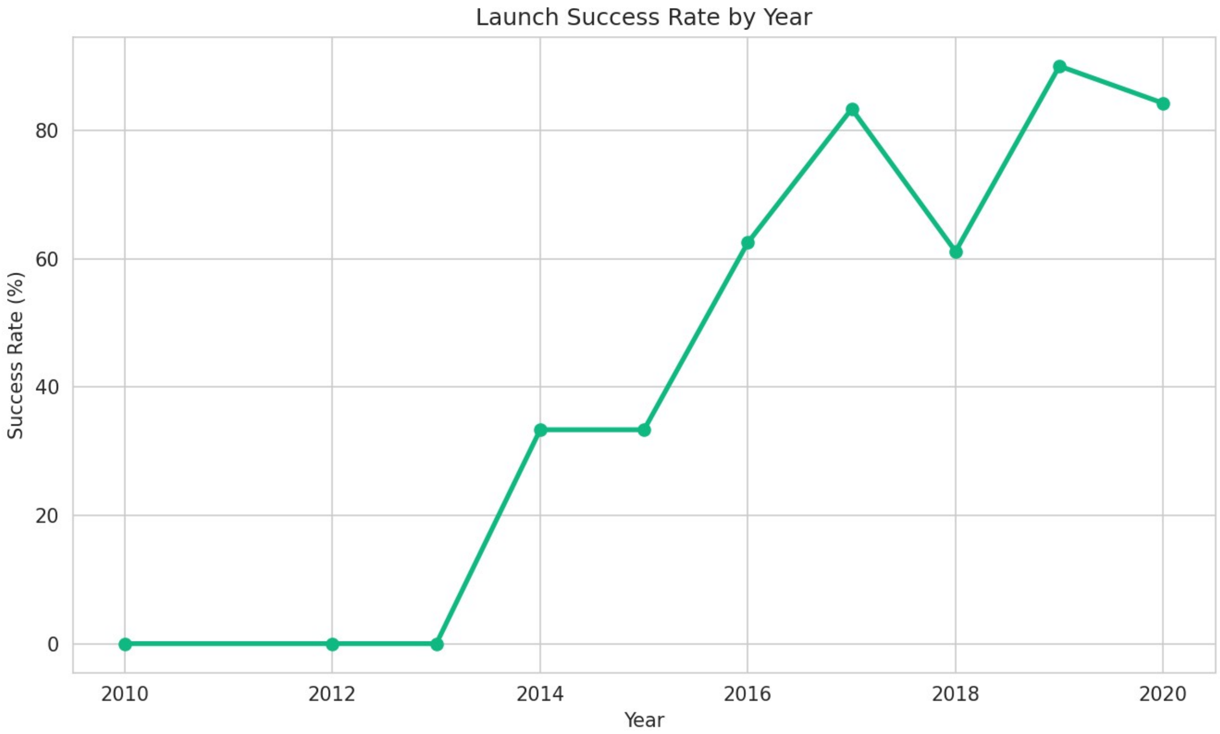
- Analysis: VAFB SLC-4E and CCAFS SLC-40 both handle a wide range of payload masses, from under 1000 kg up to 15,000+ kg, with successes and failures spread across that range.
- KSC LC-39A tends to handle heavier payloads on average and shows a higher concentration of successful (green) outcomes.
- There is no single payload threshold that guarantees success at any site - outcome depends on a combination of factors, not payload mass alone.

Result: Success Rate by Orbit Type



- Analysis: ES-L1, GEO, HEO, and SSO orbits all reach a 100% success rate, though each of these has only 1-5 launches in the dataset, so the sample size is small.
- GTO is the most common orbit by far (27 launches) but has the lowest success rate at 51.9%, making it the orbit type most in need of landing-technique improvement.
- SO (a single launch) is the only orbit with a 0% success rate.

Result: Launch Success Rate by Year



- Analysis: Success rate was 0% from 2010 to 2013, during SpaceX's earliest recovery experiments, before any successful landing had been achieved.
- A steady climb begins in 2014-2016, followed by a jump to 83.3% in 2017 and a peak of 90% in 2019.
- The dip in 2018 (61.1%) shows that progress was not perfectly linear year over year, but the overall decade-long trend is clearly upward.

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EDA with SQL - Overview

10 SQL queries were run against the wrangled dataset using SQLite, answering specific questions about launch sites, payload mass, success rates, and outcome rankings over time.

- Unique launch sites
- Records where site begins with 'CCA'
- Total & average payload mass
- Successful landings with payload between 4000-6000 kg
- Total successful vs. unsuccessful counts
- Booster(s) with maximum payload mass
- Unsuccessful landings in 2020
- Outcome ranking between two dates
- Success rate by launch site
- Average payload by orbit

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SQL Result: Launch Sites & Payload Mass

launch_site
CCAFS SLC 40
VAFB SLC 4E
KSC LC 39A

```
SELECT DISTINCT LaunchSite FROM spacex;
```

total_payload_mass	avg_payload_mass
549,446.3 kg	6,105.0 kg

```
SELECT SUM(PayloadMass), AVG(PayloadMass) FROM spacex;
```

Success Rate by Launch Site

LaunchSite	Launches	Successes	Success Rate
KSC LC 39A	22	17	77.3%
VAFB SLC 4E	13	10	76.9%
CCAFS SLC 40	55	33	60.0%

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SQL Result: Outcome Ranking (2015-2018)

Outcome	Count
True ASDS	12
True RTLS	8
None None	4
False ASDS	4
True Ocean	2
None ASDS	2

```
SELECT Outcome, COUNT(*) FROM spacex
WHERE Date BETWEEN '2015-01-01' AND '2018-01-01'
GROUP BY Outcome ORDER BY COUNT(*) DESC;
```

- True ASDS (drone-ship landing) was the most common recorded outcome in this period, at 12 launches.
- True RTLS (return to launch site) follows with 8 successful landings.
- 'None None' (no attempt / no core data) appears 4 times, alongside 4 failed drone-ship attempts.

SQL Result: Time Analysis - Unsuccessful Landings in 2020

Date	Serial	LaunchSite	Outcome
2020-01-19	B1046	KSC LC 39A	None None
2020-02-17	B1056	CCAFS SLC 40	False ASDS
2020-03-18	B1048	KSC LC 39A	False ASDS

```
SELECT Date, Serial, LaunchSite, Outcome FROM spacex
WHERE Class = 0 AND Date LIKE '2020%';
```

- Even by 2020, a small number of landing attempts still failed or had no recorded attempt, showing that landing was reliable but not perfect even late in the dataset's timeframe.

Interactive Visual Analytics - Overview

Folium Map

Launch site markers plotted with real coordinates, color-coded by landing outcome, with a calculated proximity distance from KSC LC-39A to the nearest coastline.

Plotly Dash Dashboard

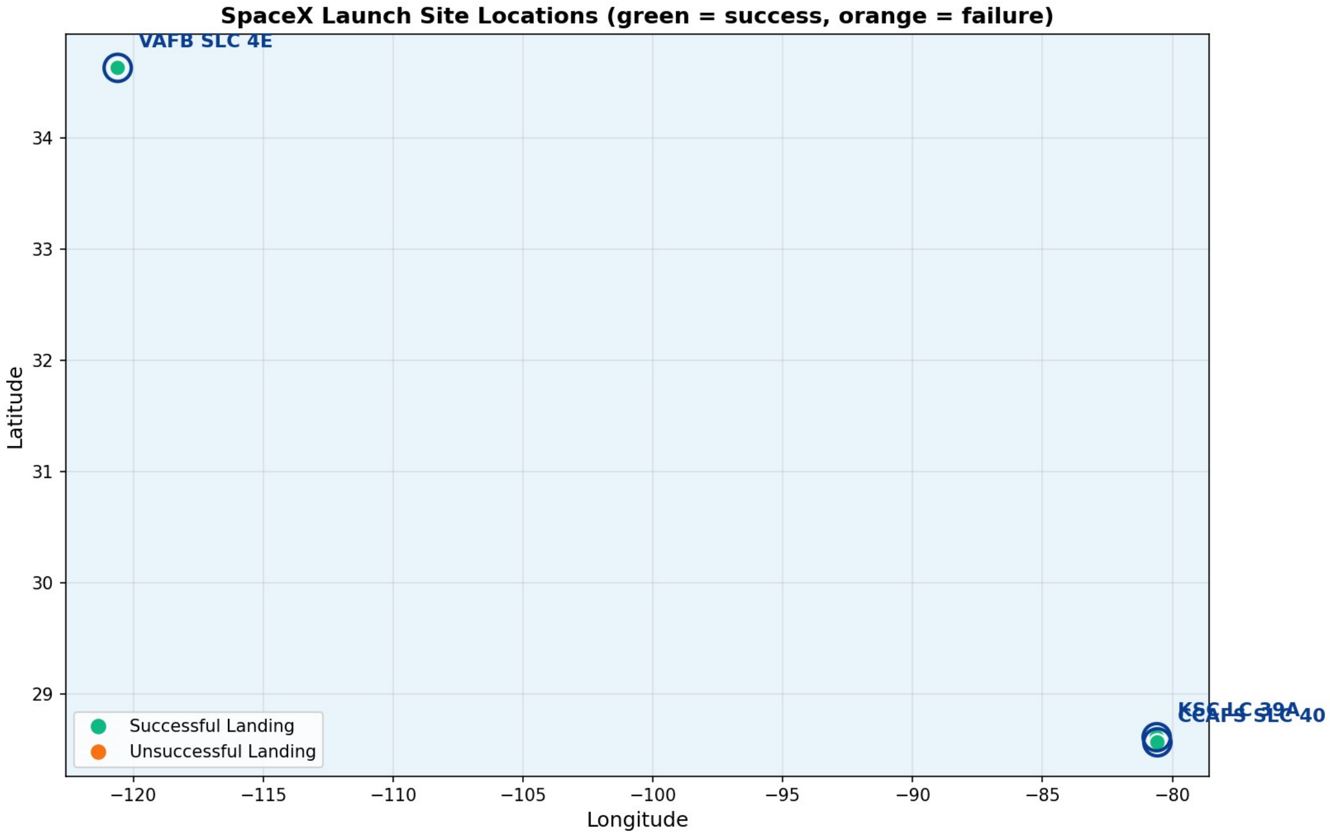
A dropdown to select a launch site, a pie chart of landing outcomes, a payload-mass range slider, and a scatter chart of payload vs. outcome.

GitHub URLs (project + notebooks):

github.com/ankitarchoudhary/IBM-Data-Science-Capstone

github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase5_Interactive_Visual_Analytics.ipynb

Result: Folium Map - Launch Site Markers



- Analysis: the map plots all 90 launches at their real latitude/longitude, color-coded green for a successful landing and orange for unsuccessful, at the three launch sites (marked with circles).
- CCAFS SLC-40 and KSC LC-39A (Florida, east coast) and VAFB SLC-4E (California, west coast) are both positioned directly on the coastline.
- Visually, KSC LC-39A shows a noticeably higher concentration of green (successful) markers relative to its total launches, matching its 77.3% success rate calculated earlier.

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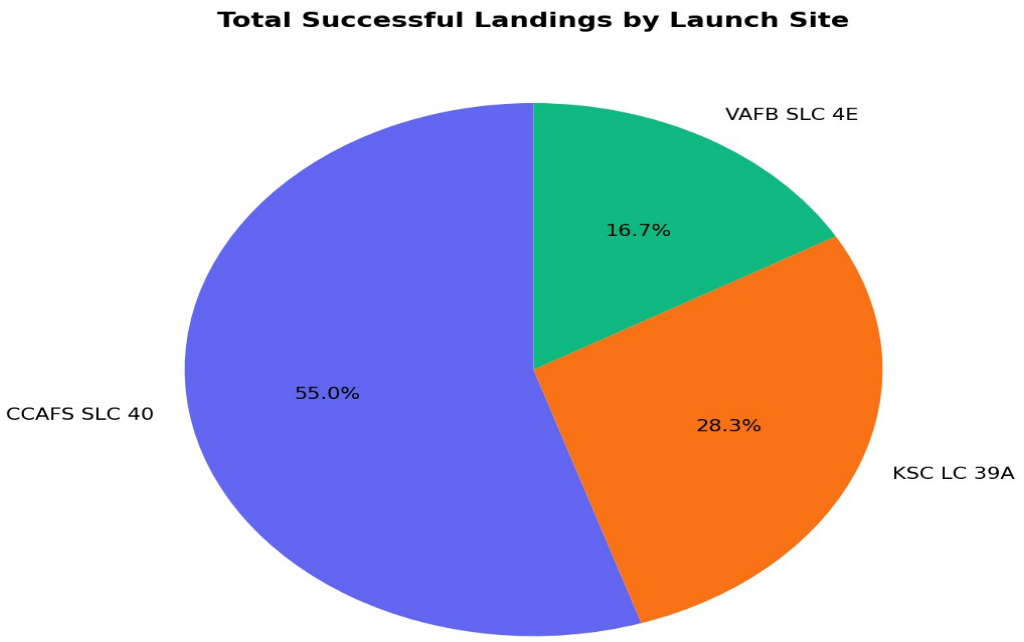
Result: Proximity Analysis

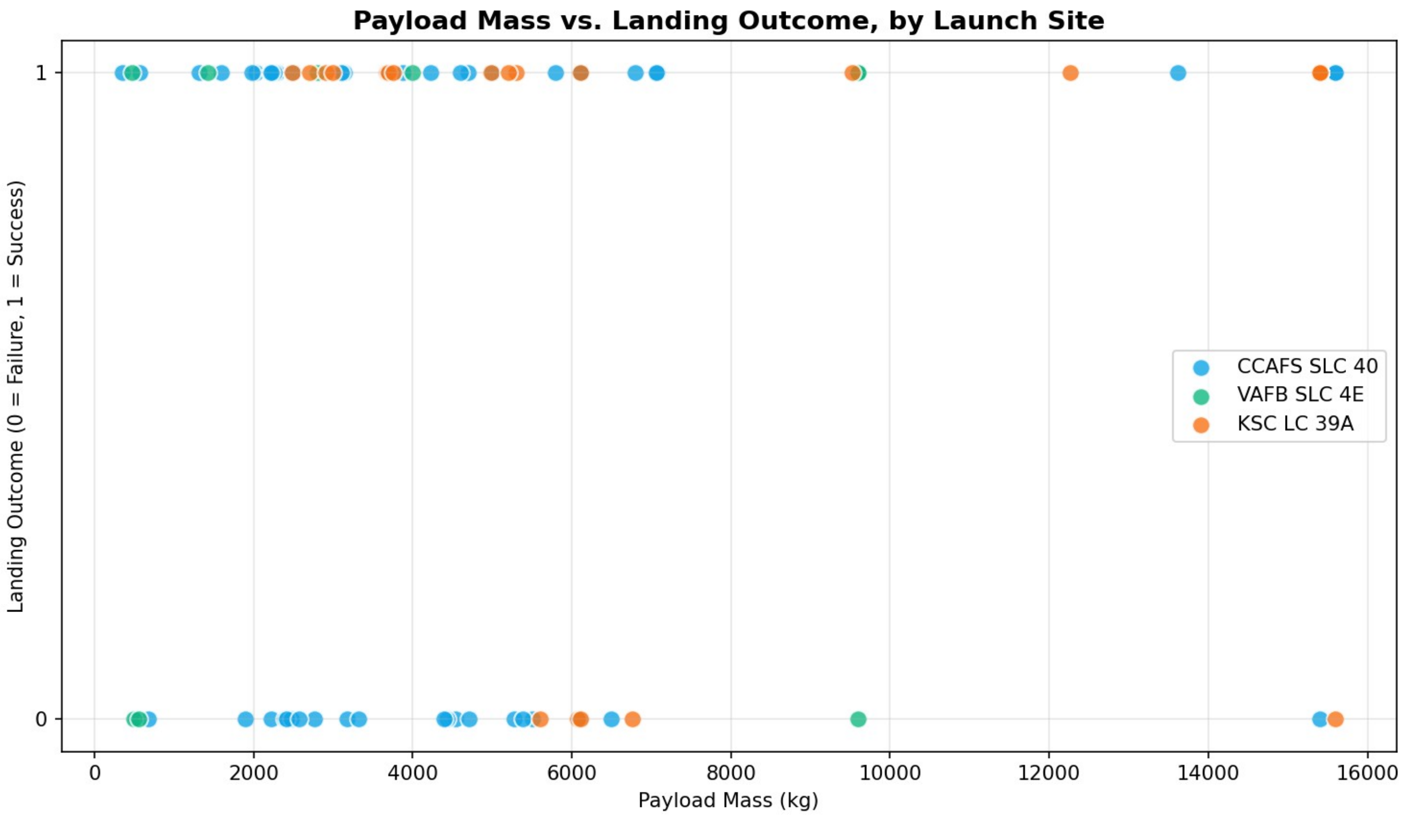
- Distance from KSC LC-39A to the nearest coastline point: 5.86 km, calculated using the Haversine formula (great-circle distance on Earth's surface).
- All launch sites are located close to the coast, so that any failed rocket falls over open water rather than a populated area.
- Most launch sites are also close to the Equator, where the Earth's surface is already moving fastest, giving an extra speed boost to anything launched from there.

Result: Dashboard - Success Pie Chart



- Analysis: this chart mirrors the dashboard's default 'All Sites' pie chart view, showing the share of total successful landings contributed by each launch site.
- CCAFS SLC-40 contributes the largest share of successful landings (33 of 60), simply because it has the most total launches, not because it has the best success rate.
- In the live dashboard, selecting a specific site from the dropdown switches this chart to show that site's own success-vs-failure split instead.

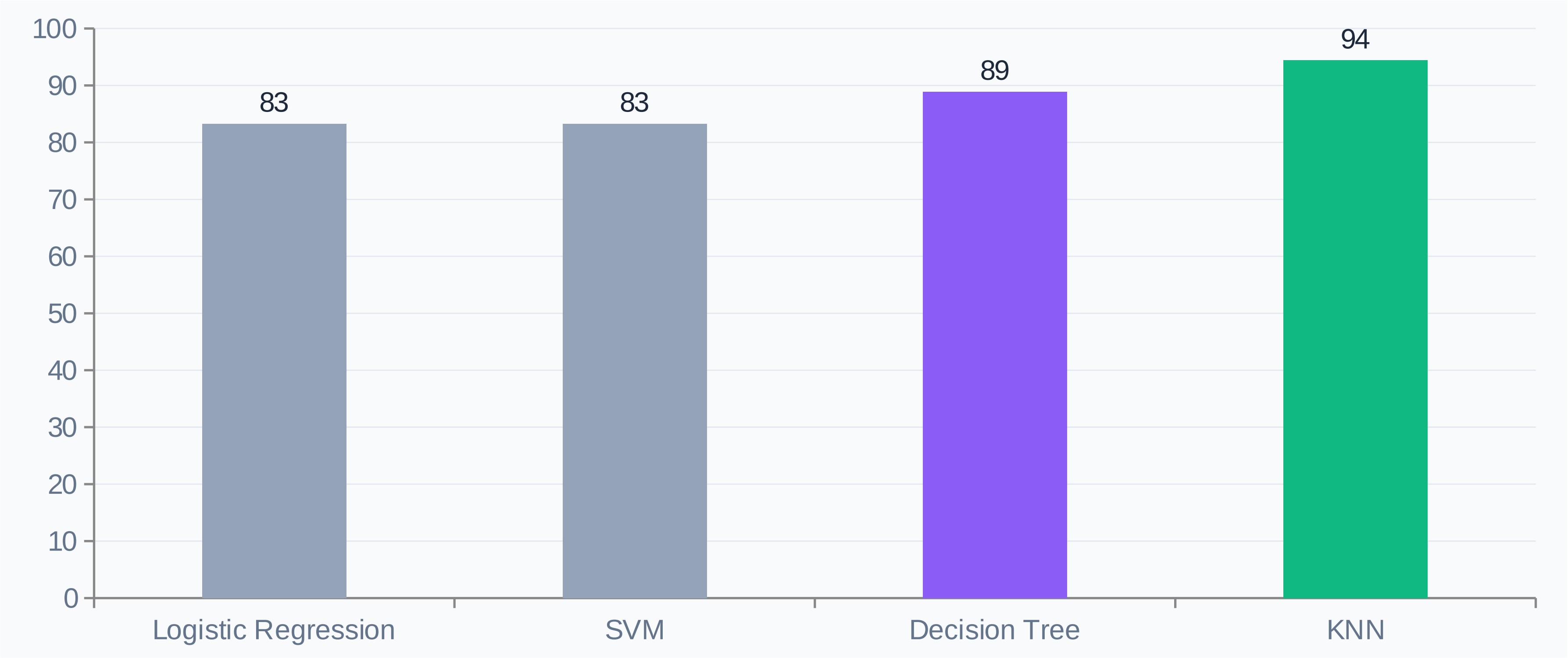
Result: Dashboard - Payload vs. Landing Outcome



- Analysis: this scatter chart mirrors the dashboard's payload-vs-outcome view, with points colored by launch site instead of booster version, since all launches in this dataset use Falcon 9.
- Successes (y=1) and failures (y=0) are visible across most of the payload range, but VAFB SLC-4E (green) launches cluster more tightly around successful outcomes.
- In the live dashboard, the range slider lets a user filter this chart down to a specific payload window interactively.

Predictive Analysis

Model Comparison - Multiple Models & Evaluation



All 4 models were tuned with GridSearchCV (5-fold cross-validation, 72 training launches) and evaluated on 18 held-out test launches. Best parameters - KNN: n_neighbors=7, Manhattan distance, uniform weights.

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Confusion Matrix - Best Model (KNN)

	Predicted: Did Not Land	Predicted: Landed
Actual: Did Not Land	5	1
Actual: Landed	0	12

Why KNN is the Best Model

- 17 of 18 test launches classified correctly (94.4% accuracy) - the highest of all 4 models.
- Zero false negatives: every actual successful landing was correctly identified.
- Only 1 false positive, a small, acceptable margin of error on a dataset this size.
- KNN likely outperformed the linear models (Logistic Regression, SVM-linear) because the relationship between features and landing success is not purely linear - it depends on specific combinations of orbit, site, and payload that KNN's neighbor-based approach captures naturally.

Conclusion

- KNN predicts Falcon 9 first-stage landing success with 94.4% test accuracy and zero false negatives, making it a strong candidate for estimating launch cost ahead of time.
- KSC LC-39A is both SpaceX's most reliable launch site (77.3% success) and, notably, was the first crewed NASA launch pad reused for commercial flight, a fact this data confirms is matched by strong technical performance.
- Orbit type reveals an innovation trade-off: GTO missions (the heaviest, most commercially valuable payloads) are also the hardest to land, at 51.9% success, suggesting future cost reductions depend on improving GTO-specific landing profiles.
- The sharp success-rate jump after 2016 lines up with SpaceX's public shift toward routine booster reuse, turning this dataset into a visible timeline of a real engineering learning curve.
- Combining SQL, visualization, and ML on the same dataset shows a consistent story from three independent angles: site, orbit, and time all point to the same conclusion, increasing confidence in the result.